

Exploratory data analysis (3)

July 9th, 2026

For academic use exclusively within the UC Davis COSMOS

Task

Understand structure

Summary statistics

Univariate distributions

Categorical analysis

Bivariate relationships

Correlation analysis

Group-wise summaries/Multivariate Visualization

Dimensionality reduction

Clustering and segmentation

Clustering visualization

Hypothesis generation

Tools & Techniques

`.info()`, `.shape`, `.columns`, `.dtypes`, `.head()`

`.describe()`, mean, median, std, IQR, min/max

Histograms, boxplots, density plots

Value counts, bar plots, pie charts

Scatter plots, grouped boxplots, violin plots, faceting

`.corr()`, heatmaps, pairplots

`groupby`, aggregation by category

PCA, t-SNE, UMAP (for visualization and structure discovery)

KMeans, DBSCAN, hierarchical clustering, Gaussian Mixture Models (GMM)

PCA/t-SNE scatter plots, cluster heatmaps, dendrograms, parallel coordinates

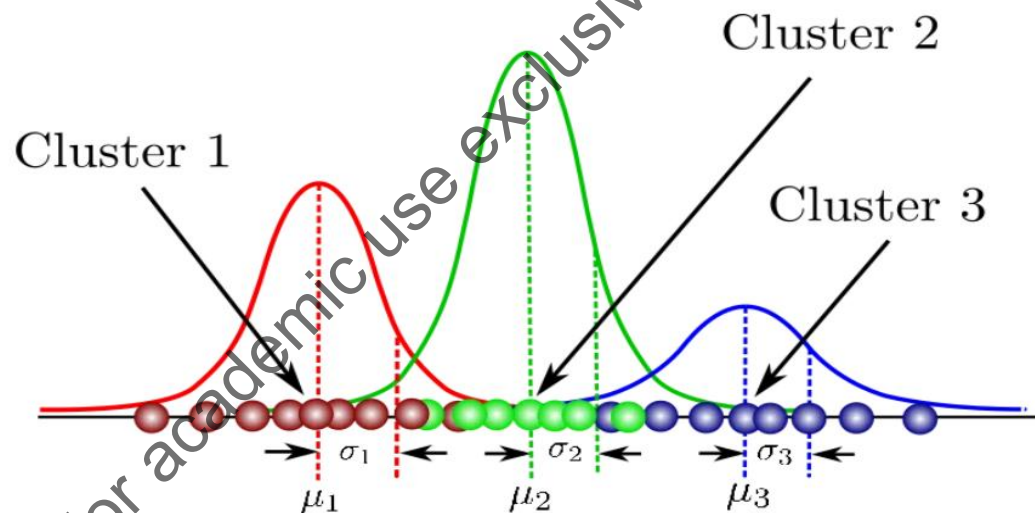
Observations and questions based on patterns or subgroup differences

Task 9: Clustering and Segmentation

- **Clustering** is the task of **grouping data points** such that those in the same group (cluster) are more similar to each other than to those in other groups.
- **Segmentation** often refers to applying clustering to real-world use cases — like customer segmentation, image segmentation, or genetic subtype discovery.

Algorithm	Type	Notes
K-Means	Centroid-based	Assumes spherical clusters, fast, widely used
Agglomerative Clustering	Hierarchical	Builds a tree of clusters, no need to pre-specify k
DBSCAN	Density-based	Finds arbitrarily shaped clusters, detects noise
Spectral Clustering	Graph-based	Good for non-convex clusters using graph Laplacian
Gaussian Mixture Models (GMM)	Probabilistic	Allows soft assignments (probabilities instead of labels)

- **K-means:** <https://www.naftaliharris.com/blog/visualizing-k-means-clustering/>
- **Agglomerative Clustering:** <https://online.stat.psu.edu/stat555/node/86/>
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise):** <https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/>
- **Spectral clustering:** Dimension reduction+K-means
- **GMM:**



Example: Real data and simulated data

- Real data: Iris dataset
- Simulated mixture of Gaussian



For academic use exclusively within the UC Davis COSMOS

Segmentation

Segmentation typically includes **3 steps**:

1.Clustering: Find groups using unsupervised learning

2.Profiling: Describe each segment using summary stats (e.g., mean income, age)

3.Application: Use segments to tailor decisions (e.g., personalized marketing)



Image segmentation



Image segmentation means dividing an image into **regions (segments)** based on similarity in pixel values, texture, or other features.

Each pixel is assigned to a **segment or object**, producing a simplified, structured version of the image.

- Load an image
- Convert it into a 2D array of RGB values
- Apply **K-Means clustering** to group pixels by color
- Reconstruct a **segmented version** using cluster labels

Task 10: Clustering Visualization



- Clustering algorithms assign data points to groups — but to **understand, trust, and refine** these results, **visualization is essential**.

Technique

2D/3D scatter plots

Cluster heatmaps

Elbow/Gap methods

Dendrograms

Purpose

Visualize clusters in reduced space (e.g., PCA, t-SNE)

Show patterns within clusters across features

Help determine the optimal number of clusters

Visualize hierarchical clustering

For academic use exclusively within the UC Davis COSMOS

Task 11: Hypothesis generation



Hypothesis generation is the process of:

- **Formulating questions or testable ideas** based on patterns, trends, or anomalies observed during EDA.
- These hypotheses can then guide further **statistical testing, model development, or experimental design.**

It's the bridge between **exploration** and **explanation.**

Dataset

Observation

Generated Hypothesis

Iris

Petal length varies by species

"Petal length differs significantly across species"

Titanic

Higher survival among
women/children

"Gender and age affect survival rate"

Customer data

One cluster has high spending

"This customer segment responds better to promos"